Titanium Track Commuter - Size 53 S - Frame Specification

Belt-drive track/commuter frame with the fit and handling geometry of the Schindelhauer Hektor (size 53), re-engineered in titanium. Rev A, 2026-10-21. All dimensions in mm and degrees; files: STEP (3D), DXF 1:1 and PDF (2D).



1. Geometry

Parameter	Hektor 53 (published)	This frame	Note
Stack	498	498.0	kept
Reach	406	406.0	kept
Head tube length	120	105.0	shortened for the taller 35 mm-clearance fork; stack/reach/BB drop unchanged
Head tube angle	73.5	73.5	kept
Seat tube angle	75.0	75.0	kept
Seat tube c-t	530	530.0	BB centre to top of seat tube
Top tube (horizontal)	530	539.4	follows from stack/reach/STA; published value is ~9 mm short in every size
Chainstay	378	395-409 (nom. 401.5)	horizontal slot; minimum set by 35 mm tyre to round seat tube
Wheelbase	961	978-992 (nom. 985)	longer than Hektor because of the longer chainstay; front centre unchanged
Standover	793	791	35 mm tyre, at BB
BB drop	n/p	48.3	derived (not published); confirmed from product photo
BB height	n/p	297.7	35 mm tyre
Fork axle-to-crown	n/p	380.0	design value incl. lower headset; must clear 35 mm tyres
Fork offset	n/p	45.0	design value
Trail	n/p	55.6	35 mm tyre

Parameter	Hektor 53 (published)	This frame	Note
Front centre	n/p	588.1	toe overlap ~28 mm w/ 170 cranks
Headset	1 1/8"	ZS44 / ZS56 tapered	Hektor uses integrated tapered 1 1/8-1.5
Seatpost	27.2	27.2	reamed Ti sleeve in 31.8 seat tube

2. Rear axle position (sliding dropouts)

Chainstay	Wheelbase	Seat-tube / tyre gap (35 mm)
395.0	978.2	4.3
401.5	984.7	10.6
409.0	992.3	17.9

Belt centre distance with a 115T belt: 401.5 mm (60/22 freewheel), 404.0 mm (60/21 fixed). Both sit inside the slot.

Clearance checks (3D model, mm)

Design tyre 35-622 (torus, 35 mm tall). Targets: tyre 4 mm, belt/sprockets 3 mm, disc rotor/caliper 3 mm. 'min'/'max' = axle at front/rear of the slot.

Check	Gap
tyre to chainstay min	6.0
tyre to seatstay min	17.2
tyre to seat tube min	4.3
belt to chainstay min	3.1
belt to seatstay min	6.5
tyre to chainstay max	6.0
tyre to seatstay max	18.9
tyre to seat tube max	17.9
belt to chainstay max	3.1
belt to seatstay max	6.5
front tyre to down tube	14.0
front crown space above tyre	34.0
disc rotor to frame min	5.2
disc caliper to frame min	4.9
tyre solid to frame min	4.3
disc rotor to frame max	6.0
disc caliper to frame max	13.0
tyre solid to frame max	5.9
tyre ok	True
belt ok	True
disc ok	True

3. Tube set (proposal - builder to confirm)

Member	Material	Size (OD x wall)	Note
Top tube	3Al-2.5V CWSR	31.8 x 0.9	straight gauge; builder may butt 0.9/0.6/0.9
Down tube	3Al-2.5V CWSR	44.5 x 0.9	oversize for BB/belt stiffness
Seat tube	3Al-2.5V CWSR	31.8 x 0.9	120 mm internal sleeve reamed to 27.2; rear clamp slot + relief hole
Head tube	6Al-4V machined	OD 48/62 tapered	ZS44 top / ZS56 bottom bearing bores (SHIS)
BB shell	6Al-4V	OD 41 x 68 wide	BSA 1.37" x 24 TPI, face & chase
Chainstays	3Al-2.5V CWSR	22.2 x 0.9	crimped to 16 x 27 oval for first 110 mm, then flared
Seat stays	3Al-2.5V CWSR	16.0 x 0.9	bent: straight 85 mm at dropout (belt clearance), then to seat tube
Dropouts	6Al-4V plate	6 mm	receivers for sliding dropouts: window + 2 x M6 slots; sliders bought in
Hose guides	Ti bosses	M4	2 on non-drive seat stay
Belt splitter	Ti / builder std.	-	bolted split in drive-side seat stay

4. Interfaces and standards

Item	Specification
Wheels / tyres	700c (622). Clearance for 35-622 tyres with at least 4 mm all round (checked in 3D). Stock Hektor: 25-622.
Rear hub	135 mm OLD disc hub (6-bolt), fixed/flip-flop or singlespeed freewheel, 10 mm bolt-on or QR axle in sliding dropouts. Rear sprocket must sit at the chainline below.
Sliding dropouts	Bolt-on sliders (e.g. Paragon Machine Works-style) in 6Al-4V receiver plates; 14 mm travel tensions the belt. Builder sets the receiver to the chosen slider system's drawing.
Drivetrain	Gates Carbon Drive CDX, 60T front, 22T (9-spline freewheel) or 21T (track thread fixed), 115T belt, chainline 50.0 mm.
Frame split	Required for belt: bolted splitter in drive-side seat stay (or split dropout).
Bottom bracket	BSA / English threaded 68 mm, external cups (stock: Hollow-Axle 170 mm cranks).
Headset / fork	ZS44/ZS56 tapered head tube; any 1 1/8"-1.5" tapered carbon fork, 380 mm A-C, 45 mm offset, 12 x 100 mm thru-axle, flat mount, 160 mm rotor, clearance for 35 mm tyres.
Seatpost / clamp	27.2 mm post; 31.8 mm external seat collar.
Brakes	Hydraulic disc. Rear: post mount, 160 mm rotor (on the non-drive slider), 160 mm rotor; caliper sits behind/above the axle and moves with the slider, so belt tensioning never needs a brake re-adjustment. Front: flat mount, 160 mm rotor on the fork.
Hose routing	Rear hose internal through top tube (6 mm ports 45 mm from each end, non-drive side), then along the non-drive seat stay on 2 x M4 guide bosses.
Mounts	2 bottle bosses on down tube, 64 mm spacing, lower at 160 mm from BB centre.

5. Manufacturing notes

- TIG weld with full back-purge (argon); no discoloured welds.

- Post-weld: face & chase BB, face/ream head tube bores to ZS44/ZS56, ream seat tube to 27.2.

- Rear receiver plates must be parallel and coplanar (belt drive and disc brake are both intolerant of misalignment): check rear triangle alignment to +/-0.5 mm and plate faces parallel within 0.2 mm; fit the actual sliders in the fixture when welding.

- Belt line: rear cog and front ring must align within 0.5 mm at 50 mm chainline - confirm with the actual hub/crank before welding the chainstays.

- Finish: brushed or bead-blasted raw titanium (no paint). No branding of Schindelhauer.

6. Open items to agree with the builder

- Builder to confirm tube gauges/butting and the chainstay crimp (or substitute a machined yoke).

- Confirm a 135 mm hub, crank and Gates sprockets that give the 50 mm chainline; the chainstay/belt gaps are designed around it.

- Choose the slider system (and whether its post mount suits a 160 mm rotor directly); the receiver plates are drawn generically and follow that system's dimensions.

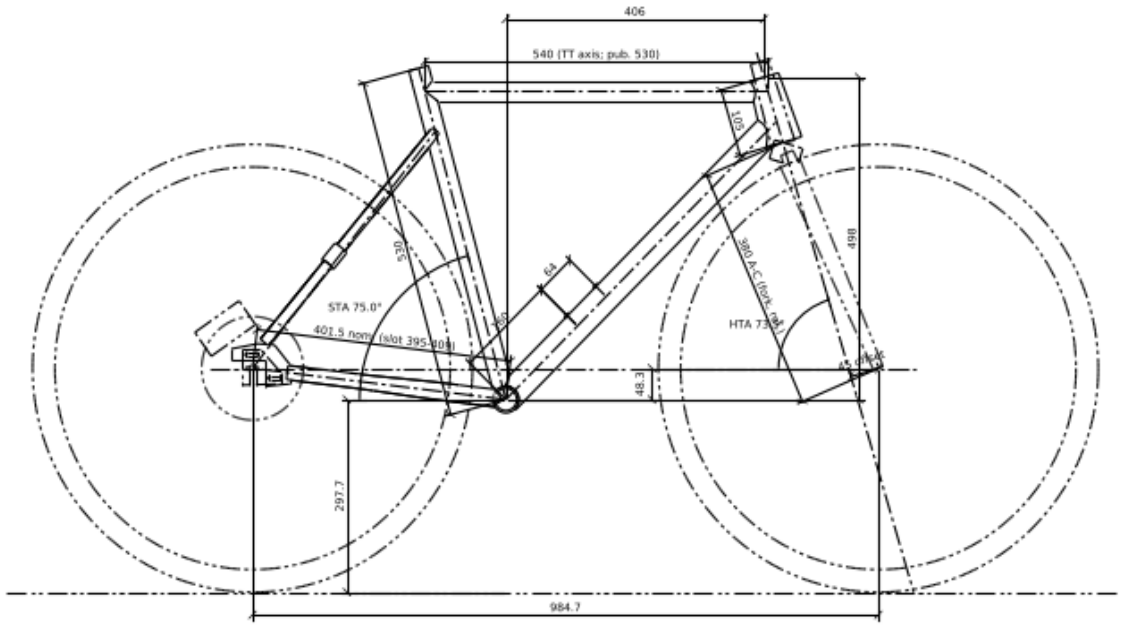
- Fork: tapered, 380 A-C, 45 offset, 12 x 100 mm thru-axle, flat mount, 160 mm rotor, 35 mm tyre clearance.

- Pick the splitter style the builder prefers (seat-stay coupler vs. split dropout).

- Optional for a commuter: mudguard eyelets, rack mounts, dynamo routing - not included yet.

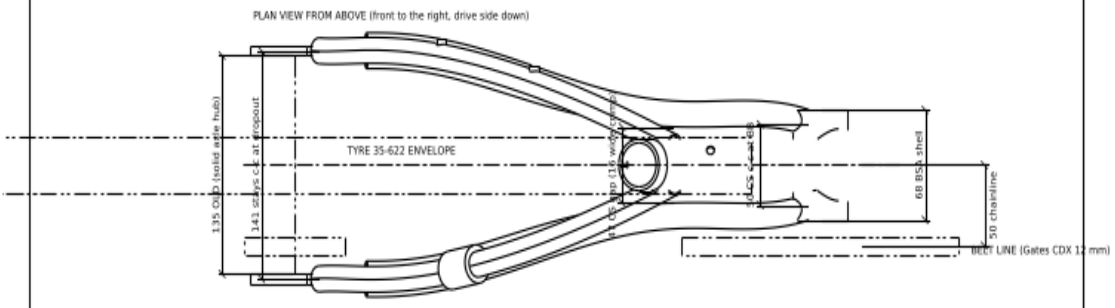
7. Published Hektor geometry, all sizes (reference)

Size	Seat tube c-t (A)	Top tube horiz. (B)	Head tube (C)	Chainstay (D)	Head angle (E)	Seat angle (F)	Wheelbase (G)	Standover (H)	Reach (I)	Stack (J)
53 S	530	530	120	378	73.5	75.0	961	793	406	498
56 M	560	550	140	378	73.0	74.5	982	816	416	516
59 L	590	570	160	378	74.0	74.0	988	841	426	537
62 XL	620	590	180	378	74.0	74.0	1008	865	440	557

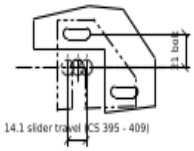


NOTES
1. Dimensions to tube centre lines; 700c (622) wheels, clearance for 35 mm tyres (min 4 mm gap all round, checked in 3D).
2. Fit geometry = published Hektor S3 (stack 498, reach 406, HTA 73.5, STA 75.0). Head tube 105 (pub. 120) for a 380/45 fork; BB drop 48.3 as Hektor.3. Horizontal TT measures 539 at stack height; published 530 is inconsistent with published stack/reach/STA (same 9-10 mm in all sizes).
4. Rear axle slot: chainstay 395–409 (min set by tyre/seat tube). 60T/22T + 115T belt sits at 401.5.
5. Trail 55.6 mm. Toe overlap approx. 28 mm (170 cranks) - expected with track geometry and large tyres.
6. Brakes: hydraulic disc. Rear post mount, 160 mm rotor (on the non-drive slider); front flat mount, 160 mm rotor on a 12 x 100 mm thru-axle fork.
7. Fork, wheels, sprockets, sliders, rotor and caliper are reference only (phantom). DXF is 1:1 - do not scale the PDF.

TITANIUM TRACK COMMUTER - SIZE 53 S
SIDE ELEVATION - FRAME GEOMETRY
Ti 3Al-2.5V / 6Al-4V PROJECTION: 1st angle
SHEET 1/2 SCALE 1:5 ON A3 UNITS mm / deg REV A



SLIDING DROPOUT DETAIL (DRIVE SIDE, OUTSIDE VIEW)
Frame: 6Al-4V receiver plate, 6 mm, window + 2 x M6 slots (6.5 wide)
Slider (phantom, bought-in): axle in vertical slot; non-drive slider carries the post mount
Axle marks: CS 395 / 401.5 (115T belt) / 409



NOTES
1. Chainstays 22.2 x 0.9 crimped to 16 x 27 oval from BB to 110 mm behind BB (clears 35 mm tyre AND 60T belt ring at 50 chainline).
2. Stays run at 70.5 mm from centre plane near the dropout so the belt and rear cog clear the drive-side stays.
3. Belt drive requires a frame split: bolted splitter in the drive-side seat stay (shown), or a split dropout per builder's standard.
4. Sliding dropouts tension the belt and carry the rear post mount, 160 mm rotor (on the non-drive slider); 160 rotor on the non-drive side. Rotor and caliper clear the frame by >= 3 mm over the full travel. No rim-brake bridge.
5. Rear hose: internal through top tube (6 mm ports, non-drive side), then along the non-drive seat stay on 2 x M4 guide bosses.

TITANIUM TRACK COMMUTER - SIZE 53 S
REAR END PLAN VIEW + DROPOUT DETAIL
Ti 3Al-2.5V / 6Al-4V PROJECTION: 1st angle
SHEET 2/2 SCALE 1:2 ON A3 UNITS mm / deg REV A